Fourier Series Properties

Property	$x(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} X[k] \quad y(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} Y[k]$
Linearity	$ax(t) + by(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} aX[k] + bY[k]$
Time Shift	$x(t - t_o) \overset{FS;\,\Omega_o}{\longleftrightarrow} e^{-jk\Omega_o t_o} X[k]$
Frequency Shift	$e^{jk_o\Omega_o t} x(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} X[k - k_o]$
Scaling	$x(at) \overset{FS;\,a\Omega_o}{\longleftrightarrow} X[k]$
Differentiation in Time	$\frac{d}{dt}x(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} jk\Omega_o X[k]$
Multiplication-Convolution	$x(t)y(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} \sum_{p=-\infty}^{\infty} X[p]Y[k-p]$
Parseval's Theorem	$\frac{1}{T}\int_{<T>} \lvert x(t)\rvert^2 dt = \sum_{k=-\infty}^{\infty} \lvert X[k]\rvert^2$
Symmetry (for $x(t)$ real)	$X[-k] = X^*[k]$ $\text{Real}\{X[k]\} = \text{Real}\{X[-k]\}$ $\text{Imag}\{X[k]\} = -\text{ Imag}\{X[-k]\}$ $\lvert X[k]\rvert = \lvert X[-k]\rvert$ $\arg\{X[k]\} = -\arg\{X[-k]\}$